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Association between ambient particulate matter and respiratory health among schoolchildren in Dar es Salaam city, Tanzania: a cross-sectional study

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Abstract

Background Urbanization and industrial activities have significantly contributed to the deterioration of air quality, with ambient particulate matter ($PM_{2.5}$ and PM_{10}) posing a major public health concern in most cities of developing countries. The impact of these pollutants on respiratory health, particularly that of schoolchildren, has remained inadequately studied.

Objective This study aimed to determine the level of ambient particulate matter ($PM_{2.5}$ and PM_{10}) and its association with respiratory symptoms among school children in Dar es Salaam, Tanzania.

Methods We conducted a cross-sectional study targeting schoolchildren aged 9–16 to investigate the impact of air pollution on respiratory health. A sample of 427 children was selected through stratified random sampling, which combined primary and secondary data. Standardized questionnaires and air quality monitoring for PM exposure were used to collect data. Analysis methods included descriptive statistics, and logistic regression to elucidate the association between PM exposure and respiratory symptoms.

Results The prevalence of respiratory symptoms among the schoolchildren was notably high, with 35.1% reporting coughing, 26.7% wheezing, 31.4% breathlessness, and 28.1% severe chest illness. Significant associations were found between proximity to pollution sources (OR = 3.4, 95% CI: 2.3–5.0, $p < 0.001$ for coughing; OR = 7.0, 95% CI: 2.9–17.2, $p < 0.001$ for wheezing; OR = 3.9, 95% CI: 2.3–6.6, $p < 0.001$ for breathlessness; OR = 6.4, 95% CI: 3.3–12.0, $p < 0.001$ for severe chest illness), household smoking (OR = 1.2, 95% CI: 1.02–1.4, $p < 0.05$ for wheezing; OR = 1.2, 95% CI: 1.01–1.4, $p < 0.05$ for severe chest illness), and the use of biomass for cooking (OR = 2.4, 95% CI: 1.4–4.0, $p < 0.01$ for wheezing; OR = 2.3, 95% CI: 1.4–3.8, $p < 0.01$ for severe chest illness) with the occurrence of respiratory symptoms. Ambient PM levels, particularly $PM_{2.5}$ and PM_{10}, exceeded WHO limits in several locations. WHO recommends $PM_{2.5}$ levels below 5 $\mu g/m^3$ and PM_{10} below 15 $\mu g/m^3$, while Dar es Salaam recorded $PM_{2.5}$ between 35–50 $\mu g/m^3$ and PM_{10} between 75–100 $\mu g/m^3$, significantly higher than recommended.

Conclusion The study emphasizes the significant impact of air pollution on the respiratory health of schoolchildren in Dar es Salaam and calls for targeted interventions to reduce particulate matter exposure, particularly in high-risk urban areas of Dar es salaam, Tanzania.

Keywords Ambient particulate matter, Respiratory health, Schoolchildren

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Introduction

Air pollution remains a critical concern globally, particularly in urban settings where vehicular emissions and industrial activities significantly contribute to the rise of ambient particulate matter. Among the pollutants, fine particulate matter ($PM_{2.5}$) and Course particulate matter (PM_{10}) pose the greatest risk to human health, as these particles can penetrate deep into the respiratory system, exacerbating conditions such as asthma, chronic bronchitis, and even mortality [14, 15, 18, 29]. Urbanization, coupled with inadequate emission control measures, has worsened the situation, especially in rapidly developing cities like Dar es Salaam [17, 20, 22, 26].

Urban residents are constantly exposed to significant levels of PM from industrial discharges, vehicular emissions, construction dust, and residential heating [13] Vehicle emissions, industrial processes, and burning fossil fuels are the sources of $PM_{2.5}$. PM_{10}, although slightly larger, also causes important health effects such as mortality, exacerbation of asthma, and chronic bronchitis. [4, 12]. Studies such as [17, 20] have consistently emphasized the role of urbanization in worsening PM levels.

In such urban areas, children are particularly vulnerable to the harmful effects of air pollution due to their developing respiratory systems and higher exposure rates during physical activities. Research has consistently shown a strong association between exposure to particulate matter and respiratory conditions in children, including reduced lung function and increased prevalence of diseases such as asthma and bronchitis [3, 14, 21, 27]. Studies specific to Dar es Salaam highlight how traffic congestion and industrial emissions drive high concentrations of $PM_{2.5}$ and PM_{10}, which often exceed the recommended limits set by the World Health Organization [20, 30–32].

The health effects of these pollutants are not limited to immediate physical symptoms but can also affect cognitive development and long-term academic performance in children [9, 24]. Given these risks, there is a pressing need for targeted public health interventions and policies aimed at reducing exposure to ambient particulate matter, especially in areas with heavy traffic and industrial activities [7, 19].

Therefore, we addressed the knowledge gap regarding the cumulative effects of PM exposure on the respiratory health of school children in Dar es Salaam. The research aims to provide critical insights for policymakers, public health officials, and educational institutions. Ultimately, these findings will contribute to the development of strategies to mitigate air pollution and protect vulnerable populations from its harmful effects.

Methods

Study design and location

A cross-sectional analytical study was conducted, utilizing both primary and secondary data collection. The study design allowed for the assessment of the association between exposure to ambient particulate matter ($PM_{2.5}$ and PM_{10}) and respiratory symptoms in schoolchildren. This study was conducted in Dar es Salaam, Tanzania's largest city, with a population exceeding six million. The city is characterized by rapid urbanization and significant environmental challenges, particularly air quality degradation caused by heavy traffic congestion and industrial activity. The study focused on five schools located in Ilala, Temeke, Ubungo, and Kigamboni municipalities, selected for their proximity to high-pollution areas within 5 km of major roads and industrial sites. The target population comprised schoolchildren aged 9 to 16 years enrolled in the selected primary schools in Dar es Salaam. This age group was chosen due to their vulnerability to air pollution and developing respiratory systems.

> *"Children with pre-existing chronic respiratory conditions, including asthma and other diagnosed long-term respiratory illnesses, were excluded from the study to minimize confounding effects. This ensured that reported respiratory symptoms were more likely attributable to ambient air pollution exposure."*

Sample size and data collection

The sample size was calculated using the Kish and Leslie formula, assuming a 50% prevalence of respiratory symptoms, yielding a minimum sample of 384 schoolchildren. A 10% non-response rate was considered, resulting in a final sample size of 427. Stratified random sampling was employed, with the number of students selected from each school proportional to its size. The schools included Amani Primary School, Diamond Primary School, Faith Primary School, Mlimani City School, and Vijibweni Primary School.

A standardized questionnaire adapted from the American Thoracic Society (ATL-DLD-78-A) was used to collect information on respiratory symptoms, medical history, and exposure to environmental pollutants.

The modified version of the ATS-DLD-78-A questionnaire was used, ensuring that only age appropriate respiratory health questions were included. Occupational exposure and emphysema-related questions were excluded.

The questionnaire was administered digitally within the selected schools. In addition, real-time air quality data on $PM_{2.5}$ and PM_{10} were collected using PurpleAir sensors installed across Dar es Salaam. These sensors provided continuous air quality measurements at two-minute

intervals, and the data was accessed via the PurpleAir online platform.

Data management and analysis

Collected data was securely stored in digital format, and pre-processing steps were taken to clean and filter the data, ensuring accuracy. The Purple Air sensors used in this study are low-cost sensors that measure particulate matter ($PM_{2.5}$ and PM_{10}) using laser particle counters. These sensors have been validated against industry standards, ensuring reliability in measuring air quality parameters. Descriptive statistics, logistic regression, and geospatial mapping were employed to explore the relationship between PM exposure and respiratory symptoms. Data analysis was performed using Stata Version 17.

Ethical considerations

This study was conducted in accordance with the ethical principles of the Declaration of Helsinki (2013 revision). Ethical approval for the study was obtained from the MUHAS Ethics Review Board. Informed consent was obtained from the guardians of the schoolchildren, and participation was voluntary. Confidentiality and anonymity of participants were maintained, and all collected data was stored securely, with access restricted to the study team.

Results

This study aimed to assess the impact of ambient particulate matter ($PM_{2.5}$ and PM_{10}) on the respiratory health of schoolchildren in Dar es Salaam.

Socio-demographic characteristics

The study included 427 participants from five schools in Dar es Salaam. The mean age was 11.5 years (SD = 2.1), with a majority of females (55.5%) and the majority of children in Standard VII (51.1%). More than 46% of the participants were over 12 years old, as summarized in Table 1.

Prevalence of respiratory symptoms

The prevalence of respiratory symptoms among the schoolchildren was notably high. The most frequently reported symptoms were coughing (35.1%), wheezing (26.7%), breathlessness (31.4%), and severe chest illness (28.1%). These findings are detailed in Table 2.

Table 1 Socio-demographic characteristics of school children (*N* = 427 participants)

Variable	Category	n (%)
School	Amani PS (Buza)	110 (25.8)
	Diamond PS (DIT)	53 (12.4)
	Faith PS (DIT)	49 (11.5)
	Mlimani SKY City	53 (12.4)
	Vijibweni PS	58 (13.6)
	Vingunguti PS	104 (24.4)
Gender	Male	190 (44.5)
	Female	237 (55.5)
Class	Std IV	209 (48.9)
	Std VII	218 (51.1)
Age Group	< 10 years	118 (27.6)
	10–12 years	109 (25.5)
	> 12 years	200 (46.8)

Table 2 Prevalence and frequency of respiratory symptoms among primary school children (*N* = 427 participants)

Variable	Response	n (%)
Cough	No	277 (64.9)
	Yes	150 (35.1)
Frequency of Coughing	Occasionally	62 (41.3)
	Monthly	51 (34.0)
	Weekly	37 (24.7)
Wheezing	No	313 (73.3)
	Yes	114 (26.7)
Frequency of Wheezing	Occasionally	56 (49.1)
	Monthly	38 (33.3)
	Weekly	19 (16.7)
Breathlessness	No	293 (68.6)
	Yes	134 (31.4)
Frequency of Breathlessness	Occasionally	61 (45.5)
	Sometimes	34 (25.4)
	Often	40 (29.9)
Severe Chest Illness	No	307 (71.9)
	Yes	120 (28.1)
Frequency of Severe Chest Illness	Once	77 (64.2)
	Twice	34 (28.3)
	More than Twice	12 (10.0)

Urban factors associated with respiratory symptoms

The study explored various urban environmental factors associated with respiratory symptoms. These factors included proximity to pollution sources, household smoking, and the use of biomass for cooking. Descriptive statistics for these variables are presented in Table 3.

The analysis revealed significant associations between proximity to pollution, household smoking, and the use

Table 3 Reported RISK EXPOsures (*N* = 427 participants)

Variable	Response	n (%)
Proximity to pollution	No	304 (71.2)
	Yes	123 (28.8)
Notice pollution in the living area	No	301 (70.5)
	Yes	126 (29.5)
Household smoking	No	362 (84.8)
	Yes	65 (15.2)
Biomass cooking	No	224 (52.5)
	Yes	203 (47.5)

of biomass for cooking, and the prevalence of respiratory symptoms. Table 4 shows the association between these factors and respiratory symptoms.

Levels of ambient particulate matter (PM2.5 and PM10)

The levels of ambient particulate matter (PM2.5 and PM10) were monitored at different locations across Dar es Salaam over multiple years. The annual PM2.5 mean levels varied significantly between locations, as shown in Fig. 1.

The levels of PM10 were particularly high, showing significant increases across sites between 2022 and 2023. This is depicted in Fig. 2

The highest PM10 levels were recorded in Vingunguti, with an annual average exceeding WHO-recommended thresholds.

Table 4 Association between respiratory symptoms and demographic and environmental factors (*N* = 427 Participants)

Factor	n	Cough n (%)	Wheezing n (%)	Breathlessness n (%)	Severe chest illness n (%)
School					
Amani PS(Buza)	110	46 (41.8)	30 (27.3)	39 (35.5)	32 (29.1)
Diamond PS (DIT)	53	17 (32.1)	13 (24.5)	15 (28.3)	13 (24.5)
Faith PS(DIT)	49	16 (32.7)	10 (20.4)	12 (24.5)	12 (24.5)
Mlimani SKY City PS	53	13 (24.5)	12 (22.6)	13 (24.5)	11 (20.8)
Vijibweni PS(Kigamboni)	58	19 (32.8)	14 (24.1)	17 (29.3)	14 (24.1)
Vingunguti PS	104	39 (37.5)	35 (33.7)	38 (36.3)	38(36.5)
Gender					
Male	190	64 (33.7)	52 (27.4)	62 (32.6)	52 (27.4)
Female	237	86 (36.3)	62 (26.2)	72 (30.4)	68 (28.7)
Class					
Std IV	209	72 (34.5)	55 (26.3)	67 (32.1)	58 (27.8)
Std VII	218	78 (35.8)	59 (27.1)	67 (30.7)	62 (28.4)
Age group					
< 10 years	118	48 (40.7)	34 (28.8)	44 (37.3)	37 (31.4)
10-12 years	109	31 (28.4)	26 (23.9)	29 (26.6)	28 (25.7)
> 12 years	200	71 (35.5)	54 (27.0)	61 (30.5)	55 (27.5)
Proximity to pollution					
No	304	43 (14.1)	10 (3.3)	28 (9.2)	15(4.9)
Yes	123	107 (87.0)***	104 (84.6)***	106 (86.2)***	105(85.4)***
Notice pollution in the living area					
No	301	48 (16.0)	13 (4.3)	31 (10.3)	21 (7.0)
Yes	126	102 (81.0)***	101 (80.2)***	103 (81.8)***	99 (78.6)***
Household smoking					
No	362	111 (30.7)	77 (21.3)	96 (26.5)	83 (22.9)
Yes	65	39 (60.0)***	37 (56.9)***	38 (58.5)***	37 (56.9)***
Biomass cooking					
No	224	36 (16.1)	10 (4.5)	25 (11.2)	13 (5.8)
Yes	203	114 (56.2)***	104 (51.2)***	109 (53.7)***	120 (28.1)***

P-Value: *** < 0.001

Association between respiratory symptoms and ambient particulate matter

There was a significant association between the levels of PM10 and the prevalence of respiratory symptoms. Higher PM10 levels were strongly associated with increased odds of coughing, wheezing, breathlessness, and severe chest illness. These findings are presented in Table 5.

This table shows the association between PM_{10} levels and the prevalence of respiratory symptoms (cough, wheezing, breathlessness, and severe chest illness) in schoolchildren. The results indicate a significant association between higher PM10 levels and increased respiratory symptoms. In areas with "Poor" air quality (PM_{10} > 75 to 100 µg/m3), a higher percentage of children reported symptoms, all with *p*-values < 0.05.

Discussion

The findings from this study reveal important insights into the impact of ambient particulate matter ($PM_{2.5}$ and PM_{10}) on the respiratory health of schoolchildren in Dar es Salaam.

The study has also revealed significant associations between proximity to pollution sources, household smoking, and the use of biomass for cooking with the occurrence of respiratory symptoms [33].

The high prevalence of respiratory symptoms observed among schoolchildren in Dar es Salaam (coughing: 35.1%,

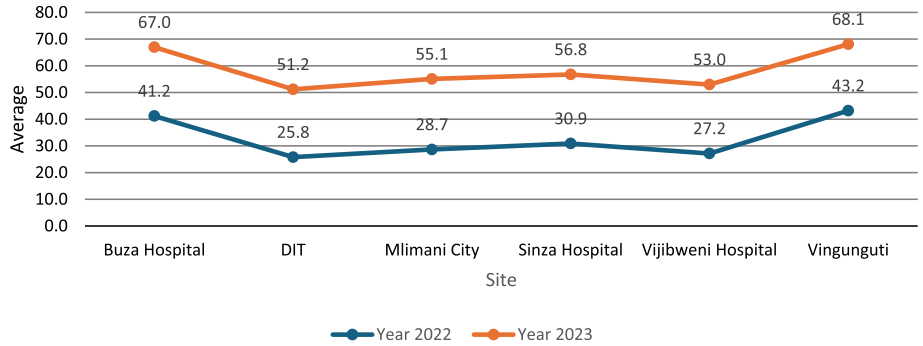


Fig. 1 Annual $PM_{2.5}$ mean comparison by site for year 2022 and 2023 (The difference in mean of each site between years was significant with p < 0.001); Level of annual mean: Good: ≤ 5 µg/m^3, Fair: > 5 to 15 µg/m^3, Poor: > 15 to 25 µg/m^3, Very Poor: > 25 µg/m^3

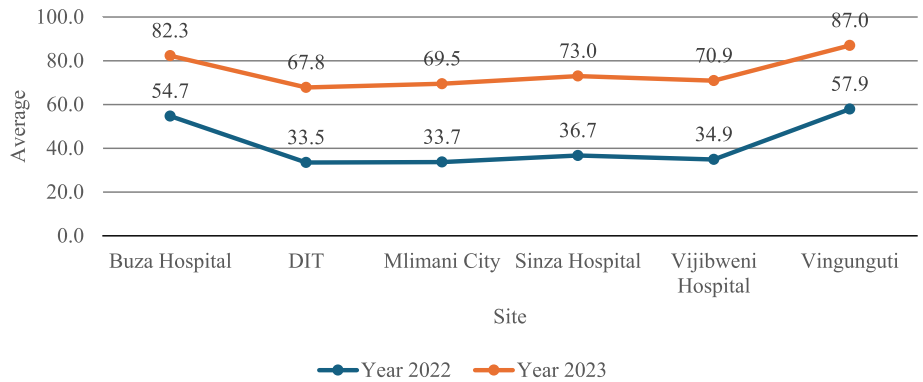


Fig. 2 Annual PM10 mean comparison by site for year 2022 and 2023 (The difference in mean of each site between years was significant with *p* < 0.001); Level of mean: Good: ≤ 45 µg/m^3, Fair: > 45 to 75 µg/m^3, Poor: > 75 to 100 µg/m^3, Very Poor: > 100 µg/m.^{3}

Table 5 Association between PM10 levels and respiratory symptoms using chi-square test

PM10 level	N	Cough n (%)	Wheezing n (%)	Breathlessness n (%)	Severe chest illness n (%)
Fair (45–75 µg/m^3)	213	65 (30.5)	49 (23.0)	57 (26.8)	50 (23.5)
Poor (> 75–100 µg/m^3)	214	85 (39.7)*	65 (30.4)*	77 (36.0)*	70 (32.7)*

P-Value: * < 0.05

wheezing: 26.7%, breathlessness: 31.4%, severe chest illness: 28.1%) aligns with previous studies that have highlighted the vulnerability of children to air pollution. These findings are consistent with research conducted by the European Respiratory Society (2023), which showed that urban children exposed to high levels of ambient particulate matter (PM) have a significantly higher risk of respiratory symptoms, particularly wheezing and breathlessness [2]. Similarly, the American Academy of Pediatrics has identified a strong connection between air pollution and respiratory health in children, linking higher pollutant levels with an increased prevalence of asthma and other chronic respiratory conditions [1, 8].

In addition, the study found that respiratory symptoms were more prevalent in children residing in areas closer to pollution sources, such as major roadways and industrial zones [6]. This supports the findings of who reported higher incidences of asthma and other respiratory symptoms among children living near busy roads [23].

The analysis revealed significant associations between urban environmental factors and the prevalence of respiratory symptoms. Specifically, children living near pollution sources, such as busy roads and industrial areas, were more likely to report respiratory symptoms. This is in line with previous studies, including those by Sierra-Vargas and Teran (2012), which emphasized the increased risk of respiratory issues among children living near high-traffic areas. Furthermore, the use of biomass for cooking and household smoking were identified as significant risk factors.

These findings are consistent with the research by the American Lung Association (2024), which showed that secondhand smoke exacerbates respiratory conditions such as asthma.

The high levels of particulate matter recorded in areas near traffic and industrial zones also highlight the critical role of these sources in contributing to air pollution [10]. A study by the European Environment Agency (2022) similarly pointed out that children are particularly vulnerable to particulate matter due to their developing respiratory systems [28].

The levels of ambient particulate matter ($PM_{2.5}$ and PM_{10}) recorded in this study consistently exceeded the World Health Organization (WHO) guidelines. This is particularly concerning as these elevated levels were associated with an increased risk of respiratory symptoms among the children [5]. The study found that PM_{10} levels in some areas reached up to 100 $\mu g/m^3$, which is significantly higher than the WHO's recommended limit of 15 $\mu g/m^3$. These findings are supported by global studies that have demonstrated the harmful effects of high PM levels on respiratory health. For instance [16], emphasized that exposure to high levels of PM10 significantly increases the risk of respiratory symptoms, particularly in urban areas with heavy traffic and industrial activity.

The logistic regression analysis showed a strong association between higher levels of particulate matter (PM_{10}) and the prevalence of respiratory symptoms, such as coughing, wheezing, breathlessness, and severe chest illness. Children living in areas with "Poor" air quality ($PM_{10} > 75$ $\mu g/m^3$) had significantly higher odds of experiencing these symptoms compared to those in areas with "Fair" air quality. This finding underscores the direct impact of air pollution on children's respiratory health, as also reported by [11, 25], who observed similar patterns in their study of children living in highly polluted urban environments.

This study provides valuable insights into the association between ambient particulate matter exposure and respiratory symptoms among schoolchildren in an urban African context, utilizing real-time air quality data and a well-validated questionnaire. The large sample size and inclusion of multiple schools enhance the generalizability of the findings.However, there are limitations, including the reliance on self-reported respiratory symptoms, which may introduce recall bias, and the inability to account for indoor air pollution or other environmental confounders. Additionally, the cross-sectional design limits the ability to infer causality between air pollution and respiratory health outcomes. As this study excluded children with known chronic respiratory conditions, findings may not be generalized to children with pre-existing respiratory diseases, who may be even more vulnerable to air pollution effects. Future studies should explore how ambient particulate matter affects children with pre-existing conditions.

Conclusion

This study highlights the significant impact of ambient particulate matter ($PM_{2.5}$ and PM_{10}) on the respiratory health of schoolchildren in Dar es Salaam, with a high prevalence of symptoms such as coughing, wheezing, and breathlessness. The strong associations between urban environmental factors, such as proximity to pollution sources and household smoking, further emphasize the vulnerability of children to air pollution in urban settings. These findings underscore the urgent need for public health interventions, stricter pollution control measures, and policies aimed at reducing exposure to harmful pollutants, particularly in high-risk areas.

To mitigate the impact of air pollution, schools should consider air purification systems, adjust outdoor activity schedules to avoid peak pollution hours, and implement community education programs to increase awareness of air pollution-related health risks.

Additionally, clinicians should take into account the environmental context when assessing pediatric respiratory symptoms in regions with high air pollution, exposure to fine particulate matter (PM2.5 and PM10) should be considered a contributing factor in respiratory distress, wheezing, and chronic cough presentations.

Future research should explore longitudinal data to better understand the long-term effects of air pollution on children's health and incorporate indoor air quality assessments for a more comprehensive understanding of the exposure risks.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12889-025-22517-6.

Supplementary Material 1.

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Authors' contributions
J.K. contributed to all aspects of the study, including conceptualization, study design, data acquisition, data analysis, manuscript writing, and review.J.B. contributed to conceptualization, data acquisition, analysis, and manuscript preparation.S.K. contributed to conceptualization, data analysis, manuscript writing, and review.A.N.M. contributed to data acquisition and conceptualization.H.S. contributed to conceptualization, data collection and manuscript preparation.All authors reviewed and approved the final manuscript.

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Data availability
The data will be available when requested from the corresponding author.

Declarations

Ethics approval and consent to participate
Ethical approval was secured from MUHAS Ethics Review Board, ensuring adherence to ethical standards in human research. Informed consent was obtained from the guardians, who are teachers at the respective schools, and the children assented to participate in the study. Participation was entirely voluntary, with the right to withdraw at any stage without penalties. Confidentiality was strictly maintained throughout the study as per ethical standards in human research.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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